Final Project Report

Project Overview

This project involved the end-to-end analysis of a large retail transaction dataset, from raw data preparation to business-focused visualization. The work was carried out in Python using pandas, Plotly, and Streamlit, and the final output is an interactive dashboard that presents key retail insights in a clear and user-friendly format.

1. Data Preparation and Cleaning

The project began with raw retail transaction data stored in CSV files. The data was inspected for structure, quality, and consistency before being transformed into a clean analytical dataset.

Key data preparation steps included:

- loading the raw transaction files,
- validating columns and data types,
- converting dates into usable datetime values,
- handling missing or inconsistent values where needed,
- preparing a cleaned dataset for analysis,
- creating an exploded dataset for product-level exploration.

2. Exploratory Data Analysis (EDA)

Exploratory analysis was performed to understand the business patterns hidden in the data. The analysis focused on:

- transaction volume and revenue distribution,
- city-wise and store-type performance,
- customer category behavior,
- seasonal trends,
- payment method usage,
- top and bottom-performing store and city combinations.

The EDA helped identify which factors most strongly influence performance and where the business is seeing stronger or weaker results.

3. Data Analysis and Metric Design

After the initial exploration, the project moved into deeper analysis to support decision-making. A set of business-oriented KPIs was created to summarize the performance of the retail business. These KPIs were designed in a way that is useful for retail and grocery-style business monitoring.

The KPI framework included metrics such as:

- total revenue,
- total transactions,
- average order value,
- unique customers,
- customer lifetime value,
- revenue trend measures over month/quarter/year,
- average items per transaction,

• discount rate,
• new and repeat customer metrics.

These KPIs were used to provide a concise executive view of the business.

4. Dashboard Development

A dashboard was built using Streamlit to present the analysis interactively. The dashboard includes:

• KPI cards for executive metrics,
• interactive filters for city and store type selection,
• charts for revenue trends and performance patterns,
• customer and payment analysis views,
• visual summaries of best and worst-performing combinations.

The dashboard was designed to be easy to use for non-technical stakeholders while still being robust enough for deeper exploration.

5. Deployment Preparation

The dashboard was prepared for deployment using Streamlit. The project files were organized so that the app can be hosted later in a Streamlit environment, including GitHub-based deployment workflows. The repository structure was arranged to support future deployment and sharing.

6. Final Outcome

The final output of this project is a complete retail analytics workflow that includes:

• cleaned and structured data,
• exploratory data analysis,
• business KPI design,
• an interactive dashboard,
• a reproducible GitHub-based project structure for sharing and deployment.

The project demonstrates how raw retail transaction data can be transformed into a practical analytics product that supports business monitoring and decision-making.